

Assignment 1

Week 1

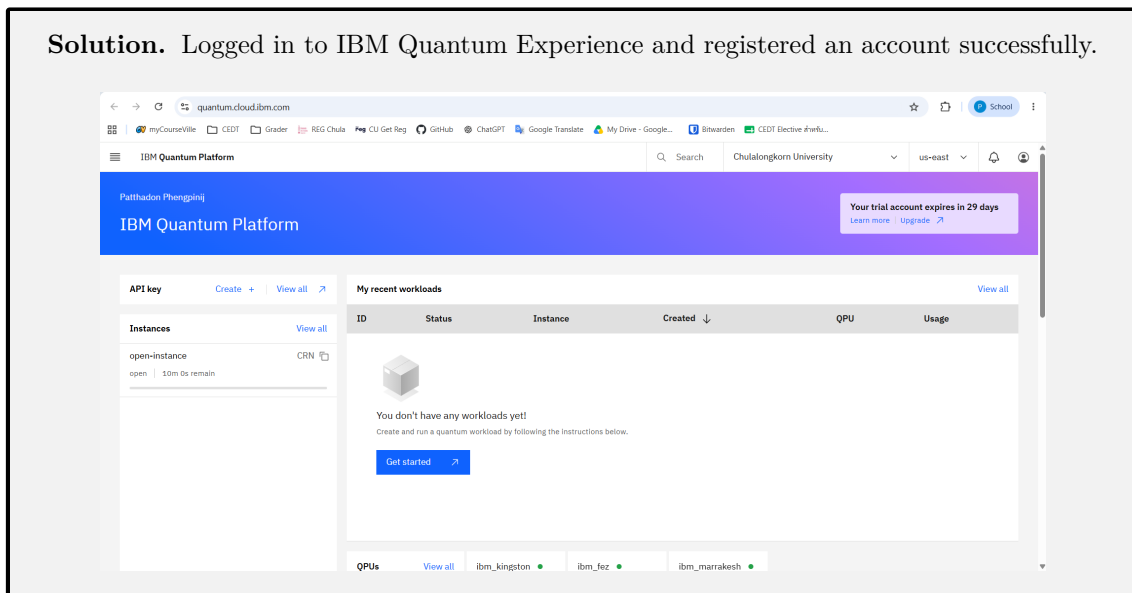
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Collaborators. Claude (for L^AT_EX styling and grammar checking)

Problem 1. Quantum Computing Account

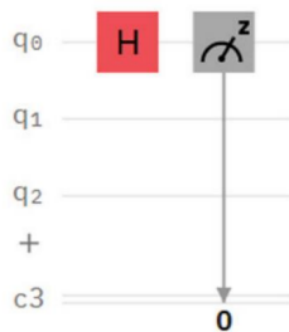
Please go to IBM Quantum Experience and register an account, <https://quantum.cloud.ibm.com/>.

Solution. Logged in to IBM Quantum Experience and registered an account successfully.



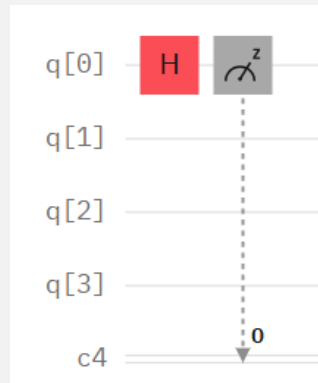
Problem 2. My First Quantum Circuit

Try to create a circuit. Click on IBM Quantum Composer and drag the gates to form the following circuit.



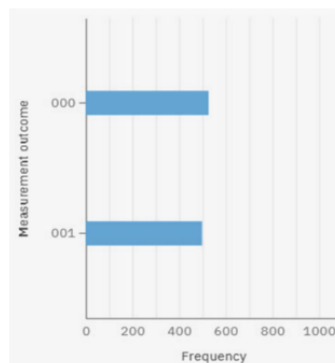
The first icon is a quantum gate called Hadamard gate and the second icon is a measurement. The input is from the left (q_0). Do not worry if you do not understand what it is at this moment. But good job! You have created your first quantum circuit!

Solution. Finished creating the quantum circuit as instructed.



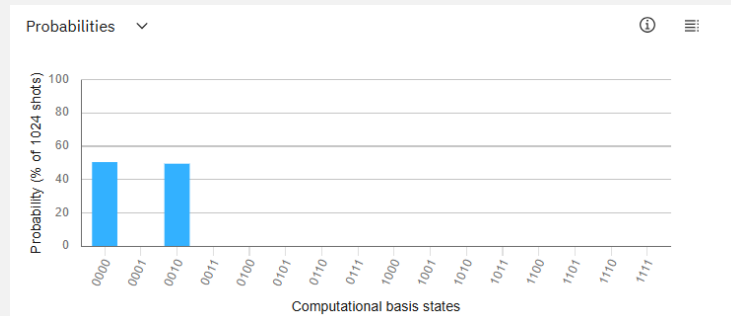
Problem 3. My First Quantum Computing

Submit the job. Click on the result when it is finished. You see some like this. You may submit to a simulator or real quantum hardware.



We started with a state in $|0\rangle$ (a basis state). After the *Hadamard gate*, it becomes a superposition of $|0\rangle$ and $|1\rangle$, i.e. $\sim |0\rangle + |1\rangle$ (we will learn later). Then we performed a measurement on the output and the superposition state collapses to either $|0\rangle$ or $|1\rangle$. The histogram shows that about 50% of the chance we obtain $|0\rangle$ and another 50% of the chance we obtain $|1\rangle$. Cool! We have done our first quantum computation. This job was submitted to a real quantum computer, called “ibmq_casablanca.” It is a 5-qubit quantum computer. However, this computer is gone. But definitely, you will find other more powerful ones.

Solution. The job was done successfully and the result was obtained as instructed.



Problem 4. Google Colab and Python

We need to do a lot of matrix operations. We will choose Google Colab, in which we can run Python easily. Since IBM Quantum Experience also uses Python, it will save your efforts. Please open a Google Colab account here <https://colab.research.google.com/>.

Solution. I decided to use my local computer to run the Jupyter notebook instead of Google Colab.

Problem 5. Vector Inner Product Using Python

We use the NumPy library to perform vector and matrix operations. To import the library, type

```
1 import numpy as np
```

Then let's implement

$$\vec{v}_1 \cdot \vec{v}_2 = (\sqrt{3} \ 1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (\sqrt{3} \times 1) + (1 \times 1) = \sqrt{3} + 1 \approx 2.732$$

```
1 v1 = np.array([np.sqrt(3), 1])
2 v2 = np.array([1, 1])
3 inner_product = np.vdot(v1, v2)
4
5 print(inner_product)
```

Type Shift-Enter to let it run. Do you see the same result? Note that we use `np.vdot` instead of `np.dot` because we need to use complex vector dot products in quantum computing. Try also `print(v1.shape)`, which will give you `(2, 1)`. It means we constructed a 2×1 matrix, which is a column vector with 2 elements.

Solution. Try running the code in a Python environment and obtained the same result of approximately 2.732.

```
1 import numpy as np
2
3 v1 = np.array([[np.sqrt(3)], [1]])
4 v2 = np.array([[1], [1]])
5 inner_product = np.vdot(v1, v2)
6
7 print(f"Inner product: {inner_product:.3f}")
8 print("v1 Shape:", v1.shape)
9 print("v2 Shape:", v2.shape)
```

Problem 6. Inner Product with Complex Coefficient 1

Now, let's try $\vec{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$. Firstly, try to use hand calculation. Then use Python to check if it is correct. Note that in the NumPy library, you need to put i as j . For example, $5 + i$ should be entered as $5 + 1j$ (putting the "1" is necessary).

Think about that before reading the code below:

```

1 v1 = np.array([[1j], [1]])
2 v2 = np.array([[-1j], [1]])
3 inner_result = np.vdot(v1, v2)
4 print(inner_result)

```

Solution. First, try to calculate the inner product manually:

$$\begin{aligned}
 \vec{v}_1 \cdot \vec{v}_2 &= \begin{pmatrix} i \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -i \\ 1 \end{pmatrix} \\
 &= (i^* \quad 1^*) \begin{pmatrix} -i \\ 1 \end{pmatrix} \\
 &= (-i \quad 1) \begin{pmatrix} -i \\ 1 \end{pmatrix} \\
 &= (-i \times -i) + (1 \times 1) \\
 &= -1 + 1 \\
 \vec{v}_1 \cdot \vec{v}_2 &= 0
 \end{aligned}$$

Then, re-check the result using Python code given by the problem. We can see that the result is also 0, which is the same as what we calculated manually.

Problem 7. Inner Product with Complex Coefficient 2

Now, let's try $\vec{v}_1 = \begin{pmatrix} 1+2i \\ 1 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$.

Solution. First, we calculate the inner product manually:

$$\begin{aligned}
 \vec{v}_1 \cdot \vec{v}_2 &= \begin{pmatrix} 1+2i \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \\
 &= ((1+2i)^* \quad 1^*) \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \\
 &= (1-2i \quad 1) \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \\
 &= (1-2i)(1-i) + (1)(1) \\
 &= 1-i-2i+2i^2+1 \\
 &= 1-3i-2+1 \\
 \vec{v}_1 \cdot \vec{v}_2 &= -3i
 \end{aligned}$$

Then, re-check the result using the following Python code:

```

1 v1 = np.array([[1 + 2j], [1]])
2 v2 = np.array([[1 - 1j], [1]])
3 inner_result = np.vdot(v1, v2)
4
5 print(f"Inner product: {inner_result:.3f}")

```

We can see that the result is also -3i, which is the same as what we calculated manually.

APPENDIX 1

Assignment 01 | Code Solution

assignment-01

August 8, 2026

1 Assignment I

1.1 Import External Libraries

```
[1]: import numpy as np
```

1.2 5. Vector Inner Product Using Python

```
[2]: v1 = np.array([[np.sqrt(3)], [1]])  
v2 = np.array([[1], [1]])  
inner_product = np.vdot(v1, v2)  
  
print(f"Inner product: {inner_product:.3f}")  
print("v1 Shape:", v1.shape)  
print("v2 Shape:", v2.shape)
```

```
Inner product: 2.732  
v1 Shape: (2, 1)  
v2 Shape: (2, 1)
```

1.3 6. Inner Product with Complex Coefficient 1

```
[3]: v1 = np.array([[1j], [1]])  
v2 = np.array([[-1j], [1]])  
inner_result = np.vdot(v1, v2)  
  
print(f"Inner product: {inner_result:.3f}")
```

```
Inner product: 0.000+0.000j
```

1.4 7. Inner Product with Complex Coefficient 2

```
[4]: v1 = np.array([[1 + 2j], [1]])  
v2 = np.array([[1 - 1j], [1]])  
inner_result = np.vdot(v1, v2)  
  
print(f"Inner product: {inner_result:.3f}")
```

Inner product: $0.000-3.000j$
